



- **Problem Statement ID** – 26187
- **Problem Statement Title** - AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.
- **Theme** - Blockchain & Cybersecurity
- **PS Category** - Software
- **Team ID** - SW-73



AI-Based Intelligent Video Analytics Platform for Border Surveillance using existing CCTV Infrastructure.



PROBLEM DESCRIPTION

- Border security forces use CCTV cameras at Border Out Posts (BOPs), checkpoints, border roads, and other important locations.
- Traditional CCTV mainly provides live video and recording.
- These systems often need continuous human monitoring, which is difficult and tiring.
- Advanced surveillance systems usually require expensive, specialized hardware and software.
- Deploying such systems across large and remote border areas can therefore be costly and difficult.

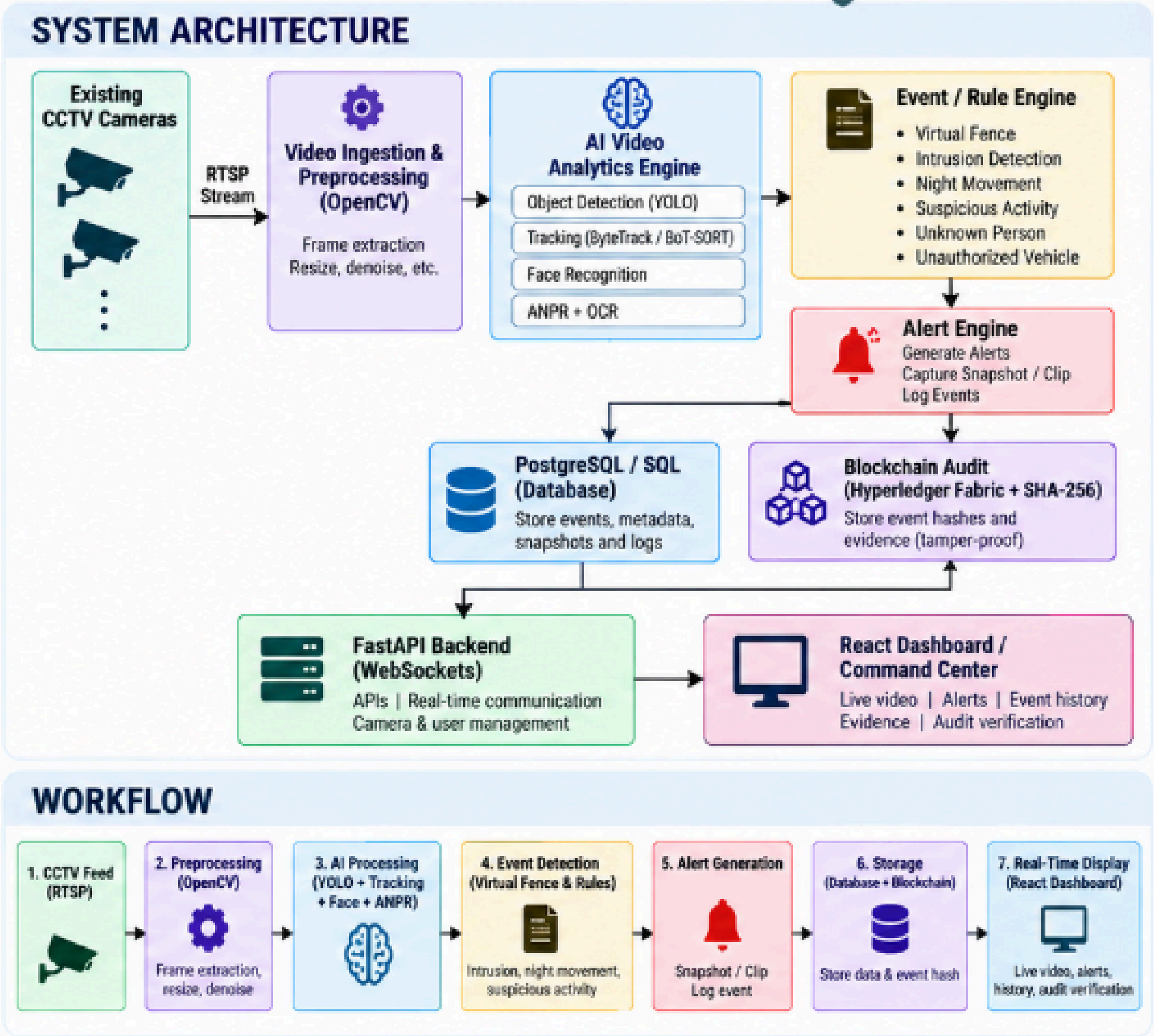
PROPOSED SOLUTION

- **Advanced AI-powered features** using existing CCTV:
 - a. Face recognition
 - b. Automatic Number Plate Recognition (ANPR)
 - c. Intrusion detection
 - d. Object tracking
- Eliminates the need for expensive, specialized hardware and software.
- More affordable and scalable surveillance system that works with existing CCTV cameras.
- Reduces dependency on continuous human monitoring and enables real-time alerts.
- **Tamper-evident security records using blockchain for critical events and evidence verification.**

TECHNICAL APPROACH



MODULE	TECHNOLOGY	PURPOSE
CCTV INPUT	RTSP + OpenCV	Receive and process existing CCTV streams
OBJECT DETECTION	YOLO	Detect people, vehicles and other objects
OBJECT TRACKING	ByteTrack / BoT-SORT	Track objects across frames and maintain IDs
INTRUSION DETECTION	Virtual Fence	Detect entry into restricted zones
NUMBER PLATE	ANPR + OCR	Detect and read vehicle number plates
FACE ANALYTICS	Face Detection + Recognition	Identify known/unknown persons
EVENT PROCESSING	Event / Rule Engine	Define and analyze suspicious activities
BACKEND	FastAPI	APIs and system coordination
DATABASE	PostgreSQL / SQL	Store events, user data, metadata and logs
BLOCKCHAIN AUDIT	Hyperledger Fabric + SHA-256	Tamper-evident audit of critical events
REAL-TIME ALERTS	WebSockets	Real-time notification to dashboard
DASHBOARD	React	Operator interface for live monitoring and analysis



FEASIBILITY AND VIABILITY



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Analysis of the Feasibility of the Idea

- Uses existing IP CCTV cameras through RTSP.
- Uses proven AI technologies like YOLO, OpenCV and OCR.
- Uses open-source tools, reducing development cost.
- Modular and scalable for multiple cameras.
- Provides real-time alerts through the dashboard.
- Cost efficient.
- Easy deployment.

Potential Challenges and Risks

- High processing requirements for multiple cameras.
- False detections/alarms due to lighting, crowds or camera angles.
- Poor footage affecting ANPR and face recognition.
- Network interruptions in CCTV streams.
- Large storage and privacy requirements.

Strategies for Overcoming These Challenges

- Use lightweight models and GPU acceleration.
- Apply confidence thresholds and tracking to reduce false alerts.
- Use image preprocessing for better OCR/face recognition.
- Add automatic stream reconnection and buffering.
- Use secure, optimized storage and role-based access.
- Start with a small number of cameras and scale gradually.

IMPACT AND BENEFITS

SOCIAL BENEFITS

- Improves public and personnel safety.
- Reduces human error and monitoring fatigue.
- Enables quicker response to suspicious activities.

ECONOMIC BENEFITS

- Uses existing CCTV infrastructure, reducing hardware costs.
- Reduces the need for manual monitoring.
- Open-source technologies help lower software costs.

POTENTIAL IMPACTS

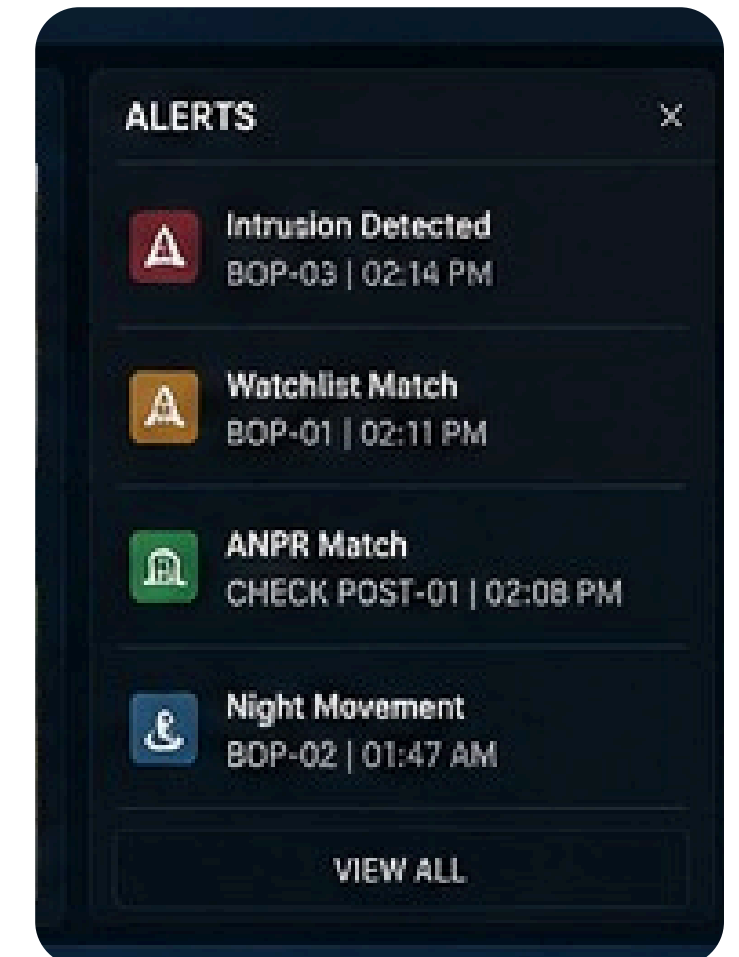
- Faster threat detection and response.
- Reduces manual CCTV monitoring.
- Improves border security and situational awareness.
- Provides real-time alerts for suspicious activities.

ENVIRONMENTAL BENEFITS

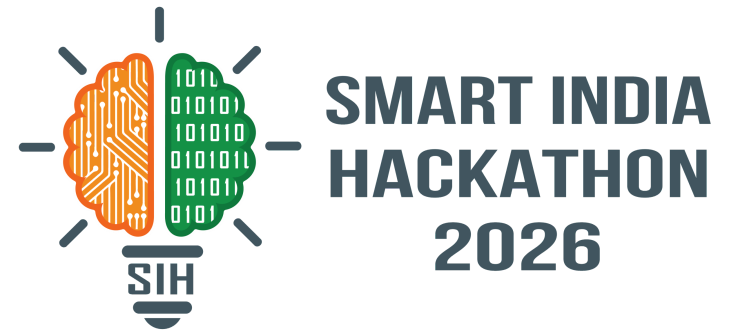
- Reduces the need for extra surveillance hardware.
- Efficient event-based storage reduces data storage and energy requirements.
- Smarter use of existing infrastructure.

OPERATIONAL BENEFITS

- 24/7 automated surveillance.
- Centralized dashboard for monitoring multiple cameras.
- Faster incident detection, verification and response.



ByteForge



THANK YOU